ALEKHA CHANDRA SAHOO

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PROFESSIONAL SUMMARY

AI-assisted software developer, systems builder, and MTech CSE student specializing in rapid prototyping, prompt-engineered development, software testing, and end-to-end cloud deployment. Leverages advanced AI tooling alongside robust verification practices to architect functional web platforms and mobile applications. Proven track record managing cloud hosting, custom domain configurations, database architecture, and live product iteration, from concept validation to production release.

EDUCATION

Master of Technology (MTech) in Computer Science & Engineering — Odisha University of Technology and Research (OUTR), Bhubaneswar *(2025 – 2027)*

Bachelor of Technology (BTech) in Computer Science & Engineering — Government College of Engineering, Kalahandi (GCEK), Bhawanipatna *(2020 – 2024)*

Higher Secondary (Science with IT) — Rajdhani Higher Secondary School, Bhubaneswar *(2018 – 2020)*

Secondary School Examination (Matriculation) — Aishanyeswar Bidyapith, Astarang *(2015 – 2018)*

CORE COMPETENCIES & TECHNICAL PROFICIENCIES

- **Development & Prototyping:** AI-Assisted Engineering, Prompt Engineering, Rapid System Prototyping, Software Testing & Iteration, Responsive Web Design, Android Architecture (Jetpack Compose, Studio)
- **Deployment & Infrastructure:** GitHub & GitHub Pages, Custom Domain Configuration & DNS Setup, Professional Email (Google Workspace), Production Deployment, SEO
- **Databases & Media Storage:** Firebase Firestore, Firebase Realtime Database, Cloudinary Media Management, Local Storage
- **Integrations & Protocols:** Razorpay Payment Gateway, WebRTC / Real-Time Media Calling, AI APIs & LLM Integration, Google Sign-In, REST APIs
- **Foundational Knowledge:** Operating Systems, Virtualization Concepts, Web Security Principles, DBMS, Data Structures & Algorithms (ADSA)
- **Programming Languages:** C, Java, Python

RESEARCH & TECHNICAL CONCEPTS

Metadata-Based Invisible Watermarking for Deepfake Media Detection

- Researched media authentication frameworks to counter synthetic visual manipulations by embedding persistent, imperceptible metadata watermarks at the point of media generation to enable downstream validation platforms and end-users to reliably verify media provenance.

Virtualization-Isolated Ingress Architecture for DDoS Mitigation

- Formulated a decoupled web server defense architecture leveraging virtualization layers to shield primary backend databases and compute infrastructure from volumetric floods, deploying an isolated inspection sandbox that runs heuristic bot-versus-human verification checks prior to proxying authenticated connections.

FEATURED PROJECTS

TrustPoints – Interpersonal Relationship & Behavioral Scoring App | *Android, Kotlin, Firebase, QR Auth*

- Engineered an Android utility that processes contacts to calculate quantified trust ratings based on user-logged positive ('green flag') and negative ('red flag') interactions.
- Designed interactive daily, weekly, and monthly statistical visualization graphs to trace behavioural trends over extended periods.
- Integrated secure Google authentication, cross-device synchronization, and QR-based web companion login for desktop data accessibility.

ZippLine – Real-Time Messaging & Audio/Video Calling Platform | *Firebase Realtime DB, Firebase Auth, WebRTC*

- Architected a communication platform enabling instant text messaging, audio calls, and video calls with low latency.

- Implemented secure Firebase user authentication, real-time message delivery tracking, online/offline status, and call connection signalling.

Traffic Intelligence & AI-Driven Analytics Dashboard | *Web Dashboard, AI APIs, Real-Time Telemetry*

- Engineered a centralized monitoring dashboard capturing visitor metrics, live user events, and navigation behaviors across self-hosted web platforms.
- Integrated an AI assistant leveraging external LLM APIs to perform conversational query answering and synthesize actionable behavioral insights directly from telemetry logs.

NextPrint – Cloud-Assisted Tokenized Document Printing Platform | *Web Platform, Razorpay, Cloud Storage*

- Deployed a document prepress platform enabling QR scanning, multi-format file uploads (PDF, DOCX, Images), and print parameter configurations (copies, color profiles).
- Integrated Razorpay payment gateway to process automated transactions and generate localized fulfillment tokens for single-click shopkeeper printing.

OUTR MTech E-Classroom & Community Hub | *Web Platform, Cloud Database, Active Production*

- Built and launched an academic dashboard actively utilized by MTech CSE students at OUTR to consolidate lecture notes, schedules, and circulars.
- Embedded the OUTR Community network, providing dedicated discussion boards, rich-media updates (images/videos), and direct communication channels.

Disclosure – Geopolitical Intelligence & Public Sentiment Portal | *JavaScript, HTML5, CSS3, Firestore*

- Published an editorial and analysis site delivering structured blogs on international relations, security affairs, and public policy with interactive live Firestore sentiment polls.

LANGUAGES

Odia (Native) · English (Professional) · Hindi (Conversational)